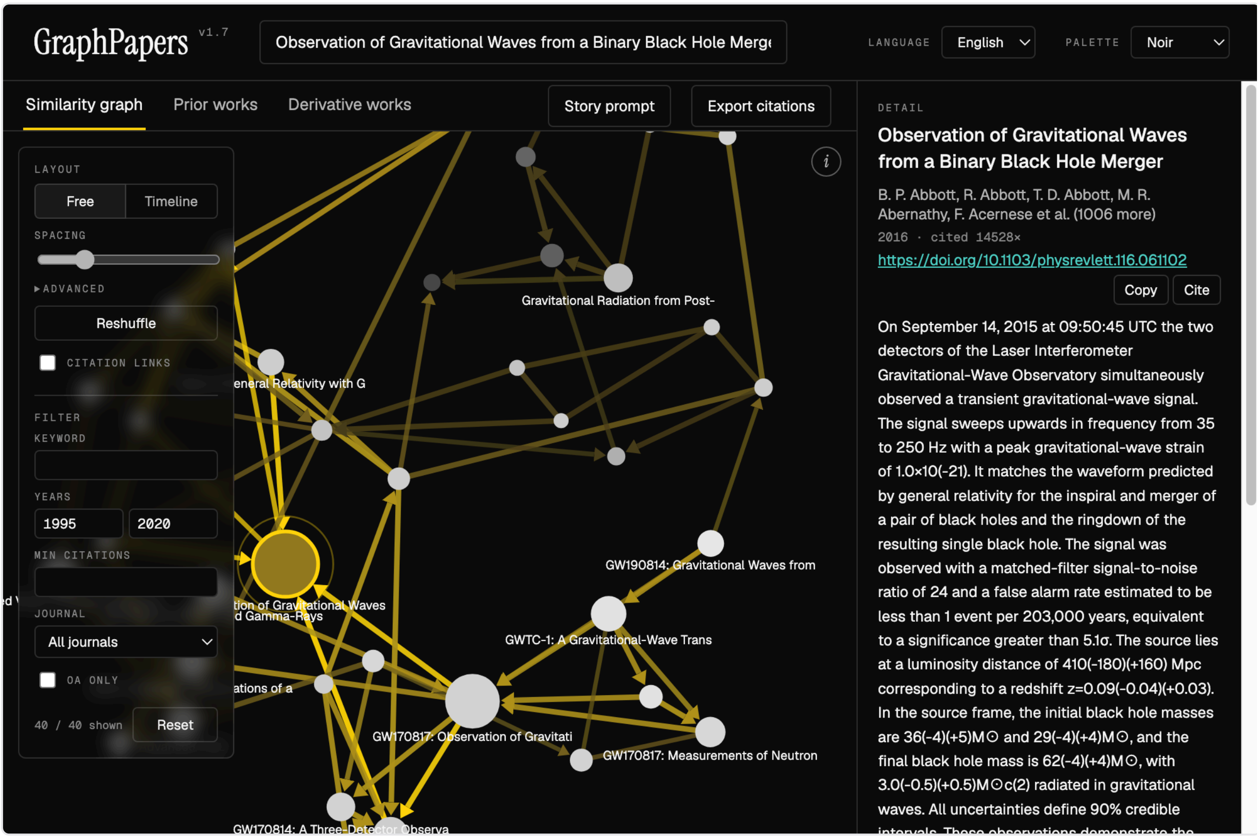

Built around the 2016 detection of GW150914. Live data, as the app always shows it. (The dark capture is one release behind — OpenAlex's daily request quota ran out before it could be retaken.)

GraphPapers

`v0.8` · Open it in a browser

A free, single-file alternative to Connected Papers. Enter a paper, get a force-directed graph of the ~40 works most related to it, then export a ready-made prompt that turns that graph into a narrative history — in your own LLM, at no cost.

No install. No account. No API key. No backend. One HTML file. Works on a phone as well as a desktop — the layout rearranges itself rather than shrinking.

Running it

Open it at **https://oddthumb.github.io/GraphPapers/**, or download `index.html` and double-click it. Either way works, and both are the whole procedure.

Everything the app needs to run is inside that file, including the d3 graph library. The only thing it fetches from the network is the paper data itself, live from OpenAlex — a free, open catalogue of 250M+ works that needs no key and imposes no cost.

Sharing it means sending a link, or one file. Either way the recipient is working immediately.

What it does

Similarity graph

Enter a title, DOI, or OpenAlex ID. The app pulls the seed paper's references and the works citing it, then scores every candidate against the seed using the same two measures Connected Papers uses:

- **Bibliographic coupling** — two papers that cite many of the same works are probably about the same thing.
- **Co-citation** — two papers that are frequently cited together are probably about the same thing.

Both measures are standard bibliometrics, not anyone's proprietary method — bibliographic coupling is Kessler (1963), co-citation is Small (1973).

The 40 highest-scoring papers become the graph. Each node connects to its three nearest neighbours, so clusters emerge on their own.

An edge means similarity, not citation. Two papers are joined because they cite the same works or are cited together — they need not cite each other at all. Where a real citation does exist between two joined papers, the edge carries an arrowhead pointing at the older work. To see citations on their own terms, tick **Citation links**: that overlays every real citation between the papers on screen, on top of the similarity layout, without moving anything.

ENCODING	MEANING
Node size	Citation count
Node colour	Publication year — pale (older) to dark (newer)
Accent ring	The seed paper
Edge weight	Similarity between the two papers
Edge colour	Distance from the seed — the accent, fading with each hop

Everything that changes the graph lives in one panel at its top-left; everything that only reads from it — **Story prompt**, **Export citations** — stays in the tab bar. The **i** button at the top-right opens the full legend.

Drag nodes, scroll to zoom, and use the **Spacing** slider to loosen or tighten the layout, or open **Advanced** for the four forces underneath it — link distance, pull, push and collision gap, each slider centred on the value the layout was tuned to. **Reshuffle** throws the nodes to new starting positions and lets the layout settle again — a force layout falls into whichever arrangement its start positions lead to, so a tangled graph often untangles on the second try. **Timeline** stacks the papers by year instead, oldest at the top, so citation arrows all point upward into the past; each year that actually occurs gets an equal band, so a single old reference cannot squash the recent decade into a sliver. Three colour palettes sit in the header.

Click a node to pin its neighbourhood; click empty canvas to release it. Click any node to see its abstract, authors, and DOI — and to rebuild the whole graph around it.

Prior and derivative works

Two tabs beside the graph list what the seed paper cites and what cites it, ranked by citation count.

Quantum Jump

Following citations keeps you inside one conversation. **Quantum Jump!** leaves it: it moves the graph to a paper that has *no citation link* to the current one, and draws a trail back to where you came from, so the path stays visible however far you wander.

Two ways of finding those papers, chosen with **Find by**:

	IT LOOKS FOR	IT CANNOT SEE
Read together	Papers other researchers cite alongside this one, though the two never cite each other	Papers too recent for anyone to have cited them yet
Shared foundations	Papers that cite the same *rare* works this one cites	Work built on a different literature
Same ideas	Papers tagged with subjects you choose, and no bibliometric tie at all	Work OpenAlex has tagged differently
Linked through a third paper	Papers with no link to yours, both strongly tied to some intermediate — Swanson's 1986 method, which connected fish oil to Raynaud's disease through blood viscosity	Pairs with no intermediate in common

Same ideas hands you the seed's own subject tags and lets you pick which to combine. The choice decides the reach: keeping the object terms holds you inside the field, while picking a measurement term — *spectral analysis*, say — reaches work in medicine or engineering doing the same kind of analysis. OpenAlex does not mark which tags are which, and only the person reading knows which axis matters.

The four routes returned no overlapping candidates at all in testing, on either seed.

Rarity is what makes the second one work. Sharing a field's standard catalogue with someone is no evidence of anything; sharing a paper cited eight times means you are looking at nearly the same problem.

Both are corrected for fame — a raw count of shared citations just returns whichever papers everyone cites. Both drop candidates already drawn, so pressing the button again gives you somewhere new; **Undo jump** steps back, and **Reset draws** starts over. **Field** narrows the search to the same discipline, or restricts it to a different one.

Hover the trail to see why two papers were linked: the works that cite them both, or the references they share.

Citation export

Export citations writes the papers out as BibTeX, RIS, APA, MLA, or Chicago — one paper, everything the filters currently show, or all 40. Copy it, or download a `.bib` / `.ris` file and drag it straight into Zotero, EndNote, or Overleaf.

Author names come from OpenAlex as single strings, so the family name is taken as everything after the last space. That is right for most Western names and wrong for some; check the names before submitting.

Filtering

In the same left-hand panel: keyword (title and authors), year range, minimum citations, open access, and journal impact. The journal filter uses OpenAlex's 2-year mean citedness, ranked among the journals present in the current graph — it is **not** a JCR quartile, which is not open data. Filtering hides papers without moving the layout, and resets whenever you build a new graph.

LLM story prompt

The **Story prompt** button assembles a prompt describing the graph — every paper grouped as prior, later, or related, with its year, first author, citation count, and distance from the centre in graph hops.

Copy it into ChatGPT, Claude, Gemini, or anything else, and you get a written history of the research line: what problem the centre paper solved, what led to it, and what it produced.

DEPTH	PAPERS	USE IT FOR
1	~10	A tight story about the paper's immediate neighbourhood
2	~25	The default — enough context to see a lineage
3	~37	The whole field around the paper
All	39	Everything in the graph

Output language is switchable between English and Korean.

The app builds this text itself — it never calls an LLM, so this step costs nothing, needs no key, and works instantly. Which model reads it is entirely your choice.

Data freshness

There is no local cache and no database. Every lookup goes straight to OpenAlex, so what you see is what OpenAlex holds at that moment. The bottom-right corner always shows the time of the current fetch and the date OpenAlex last updated the seed record, so you know exactly what you are looking at.

The cost of that guarantee is that **the app needs an internet connection**. Without one it opens and renders, but cannot fetch papers, and says so.

OpenAlex allows 1,000 requests a day, counted per IP address — your own, not shared with anyone else using GraphPapers. A graph costs four to eight requests and a Quantum Jump one to six, so the ceiling is somewhere above a hundred graphs a day and most people will never approach it. The bottom-right corner shows what you have left and when it resets, and turns the accent colour as it runs low, so the limit is visible before it is reached rather than arriving as an unexplained failure.

Requirements

A modern browser. That is all.

On a phone

The layout rearranges below 760px rather than scaling down. The graph and the detail panel stop sitting side by side and the panel becomes a sheet that slides up when you tap a node, since a strip under a full-height graph cannot be read and cannot be scrolled to — dragging over the canvas pans the graph. The control panel folds behind a button so it does not cover what it controls, and the header wraps to give the search box a full row. Pinch to zoom and drag nodes work on touch.

Google Fonts is the only other external reference; if it is unreachable the app falls back to system fonts and works normally.

Credits

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Ideas, bug reports, and pull requests are welcome; contributors are credited in the app, under the Acknowledgements link in the bottom-right corner.

Paper data from OpenAlex. Graph layout by d3.

Still 0.x: the shape of this changes often. 1.0 is reserved for when it has been used on a real manuscript and earned the right to stop moving.

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